

TITLE-PreCare AI An Agentic Artificial Intelligence Based Pregnancy Risk Analysis and Maternal Health Monitoring System

ABSTRACT

The present invention relates to an AI-powered pregnancy monitoring and healthcare assistance system designed for intelligent maternal healthcare management. The system integrates a mobile application, FastAPI backend server, OCR technology, artificial intelligence algorithms, and database management for automated healthcare support. Pregnant users can upload medical reports in image or PDF format through the application. OCR technology extracts medical information from the uploaded reports for further analysis. The extracted data is processed using AI algorithms to predict maternal health risks and generate healthcare recommendations. The system classifies pregnancy conditions into low, moderate, and high-risk categories. The platform also supports continuous health tracking, including monitoring of blood pressure, glucose levels, weight, and pregnancy symptoms. Dashboard visualization is provided for real-time health monitoring and analysis. The invention includes appointment scheduling and healthcare reminder functionalities. Emergency healthcare alerts are generated during critical maternal health conditions to improve patient safety and response time. The centralized backend infrastructure enables secure healthcare data storage and efficient communication between system modules. The invention improves early risk detection, automated report analysis, and efficient maternal healthcare management through intelligent healthcare technologies.

THE FIELD OF INVENTION (HEALTHCARE INDUSTRY)

The present invention relates to the field of healthcare technology, artificial intelligence, and medical data processing. More specifically, the invention relates to an AI-powered pregnancy monitoring and healthcare assistance system that utilizes OCR technology, machine learning algorithms, and real-time health tracking for automated medical report analysis, maternal risk prediction, emergency alert generation, and healthcare management.

BACKGROUND OF THE INVENTION

Maternal healthcare requires continuous monitoring of pregnancy-related health conditions, medical reports, and timely healthcare assistance to ensure the safety of both mother and child. Conventional pregnancy monitoring methods mainly depend on manual interpretation of medical reports, hospital visits, and periodic consultations with healthcare professionals. Such methods may lead to delays in identifying high-risk pregnancy conditions and limitations in continuous health monitoring.

Although several digital healthcare applications are available for appointment scheduling and symptom monitoring, many existing systems lack intelligent medical report analysis, automated risk prediction, and integrated emergency healthcare support. Pregnant women may experience difficulty in understanding medical reports and identifying potential complications at an early stage, which may increase healthcare risks during critical conditions.

Therefore, there is a need for an intelligent maternal healthcare system capable of automatically analyzing pregnancy reports, predicting maternal health risks, continuously monitoring health parameters, and generating emergency alerts when abnormal conditions are detected. The present invention addresses these limitations through an AI-driven healthcare platform integrating OCR-based report extraction, AI analysis, health tracking, and emergency healthcare assistance within a unified system.

SUMMARY OF THE INVENTION

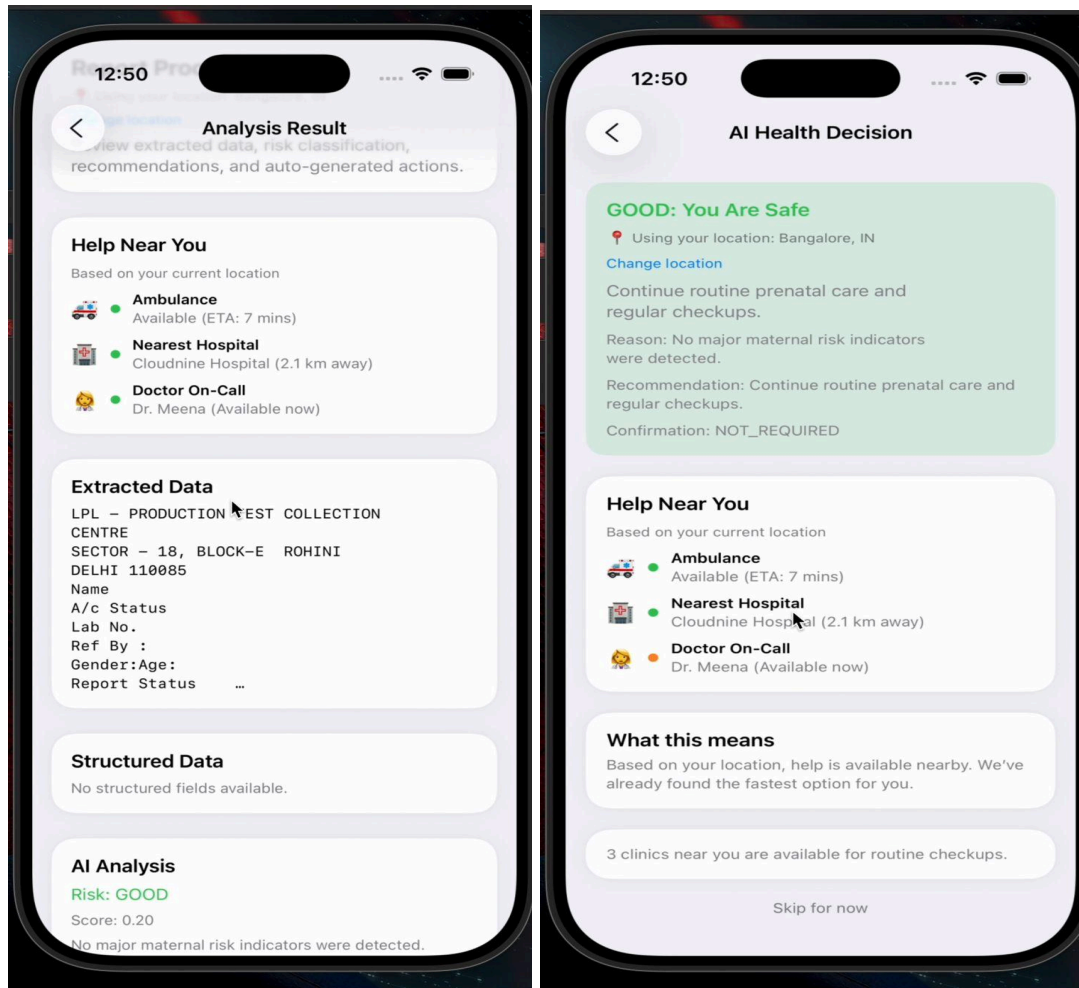
The present invention provides an AI-driven pregnancy monitoring and healthcare assistance system for intelligent maternal healthcare management. The system utilizes OCR technology to extract medical data from uploaded pregnancy reports and applies AI algorithms for automated risk prediction and healthcare analysis. The platform supports continuous health tracking, appointment scheduling, dashboard visualization, and emergency healthcare alerts through a mobile application and centralized FastAPI backend. The invention improves early risk detection, real-time healthcare monitoring, and efficient maternal healthcare management.

SPECIFICATION

- The PreCare AI Pregnancy Monitoring System is designed as an intelligent healthcare platform connecting pregnant women, healthcare professionals, and medical support services through a centralized backend infrastructure. The system integrates AI-based medical report analysis, OCR technology, health monitoring, appointment management, and emergency healthcare support to provide efficient maternal healthcare assistance.
- Users can upload pregnancy medical reports in image or PDF format using the mobile application. OCR technology extracts medical information from the uploaded reports, and AI algorithms analyze the extracted data to identify maternal health risks and generate healthcare recommendations. The system supports continuous monitoring of health parameters including blood pressure, glucose level, weight, and pregnancy symptoms.
- The platform provides dashboard visualization, appointment scheduling, healthcare recommendations, and emergency alert notifications for critical health conditions. The centralized backend infrastructure enables secure healthcare data storage, efficient communication between modules, and real-time healthcare monitoring. The modular architecture also supports scalability and future integration of additional healthcare services and intelligent monitoring features.

DESCRIPTION

The PreCare AI Pregnancy Monitoring System is an AI-powered maternal healthcare platform developed for intelligent pregnancy monitoring and healthcare assistance. The system enables users to upload pregnancy medical reports in image or PDF format through a mobile application interface. OCR technology extracts medical information from the reports, and AI algorithms process the extracted data to predict pregnancy-related health risks and generate healthcare recommendations. The platform supports continuous health tracking, including monitoring of blood pressure, glucose levels, weight, heart rate, and pregnancy symptoms. The system also includes dashboard visualization, appointment management, and real-time healthcare monitoring functionalities. During critical health conditions, the platform generates emergency alerts and healthcare assistance recommendations to improve patient safety and response time. The system operates using a FastAPI backend server and database management system for secure and efficient healthcare data processing.



WE CLAIM

1. **Claim 1:** The PreCare AI Pregnancy Monitoring System connects pregnant women and healthcare services through an intelligent healthcare platform for secure medical report analysis, maternal risk prediction, and healthcare assistance.
2. **Claim 2:** The system utilizes OCR technology and AI algorithms to extract and analyze medical information from uploaded pregnancy reports for automated healthcare assessment and recommendation generation.
3. **Claim 3:** The platform supports continuous health tracking, dashboard visualization, and real-time monitoring to improve maternal healthcare management and early risk identification.
4. **Claim 4:** The system generates emergency healthcare alerts and assistance

recommendations during critical maternal health conditions to improve patient safety and healthcare response efficiency.

5. **Claim 5:** The application includes appointment scheduling, centralized healthcare data storage, and secure healthcare record management for efficient healthcare coordination.
6. **Claim 6:** The system architecture is scalable and flexible, enabling efficient management of multiple users, healthcare records, and AI analysis processes while maintaining secure healthcare data processing.